

Lecture 23

Evolutionary Game Theory as a Dynamic View of Strategic Behaviour

COMP532 – Machine Learning and Bioinspired Optimisation

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Today's Questions

At the end of this lecture, you should be able to answer:

1. Why is static equilibrium analysis not enough for adaptive multi-agent systems?
2. What is the population view of strategic interaction?
3. What are replicator dynamics and what do they model?
4. What is an evolutionarily stable strategy (ESS)?

Recap: Where We Are in the Module

So far, we have established that:

- ▶ multi-agent learning is hard because agents adapt to one another
- ▶ game theory gives a language for strategic interaction
- ▶ equilibrium is a useful notion of stability
- ▶ learning is a dynamic process rather than a static object

Today's shift

We now move from **equilibrium as a static concept** to **behaviour as an evolving population process**.

Why Static Equilibrium is Not Enough

Nash equilibrium tells us what a stable strategic profile may look like.

But in adaptive systems, we also want to understand:

- ▶ how behaviours spread or disappear over time
- ▶ why some conventions emerge and persist
- ▶ why some systems converge and others oscillate
- ▶ what stability means under perturbation

Key point

Equilibrium describes a **state**. Evolutionary analysis describes a **process**.

What is Evolutionary Game Theory?

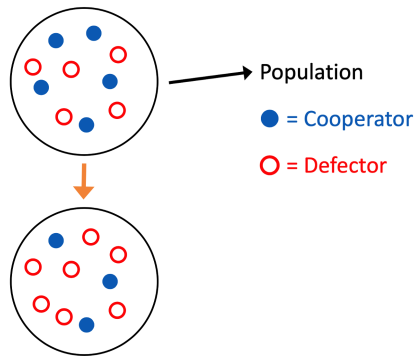
Evolutionary Game Theory (EGT) studies how the prevalence of strategies changes over time in a population.

Instead of assuming perfectly rational agents solving a game explicitly, EGT focuses on:

- ▶ populations of strategies
- ▶ relative performance
- ▶ adaptation and selection
- ▶ dynamic stability

Core idea

Strategies that perform better than average tend to become more common.



Classical Game Theory vs Evolutionary Game Theory

Classical game theory

- ▶ rational players
- ▶ explicit strategic reasoning
- ▶ best responses
- ▶ Nash equilibrium

Evolutionary game theory

- ▶ population of behaviours
- ▶ differential success
- ▶ changing strategy frequencies
- ▶ robustness under perturbation

Interpretation

Classical GT focuses on **strategic choice**. EGT focuses on **strategic evolution**.

The Population State

Suppose a population can use k strategies:

$$S = \{1, 2, \dots, k\}.$$

Let

$$x_i \geq 0, \quad \sum_{i=1}^k x_i = 1$$

denote the proportion of the population using strategy i .

Then the population state is:

$$x = (x_1, \dots, x_k).$$

Key point

A population state is a **distribution over strategies**, not a single action choice.

Example

If the two strategies are

$$S = \{\text{Cooperate}, \text{Defect}\},$$

and

$$x = (0.3, 0.7),$$

then:

- ▶ 30% cooperate
- ▶ 70% defect

Payoff as Fitness

In EGT, payoff is often interpreted as **fitness** or **growth advantage**.

If the population state is x , the payoff to strategy i is:

$$f_i(x).$$

The average payoff in the population is:

$$\bar{f}(x) = \sum_{i=1}^k x_i f_i(x).$$

Important idea

What matters is not absolute success, but whether a strategy performs **better or worse than the population average**.

Example

Suppose:

$$x = (0.3, 0.7),$$

and the current payoffs are

$$f_1(x) = 2, \quad f_2(x) = 1.$$

Then

$$\bar{f}(x) = 0.3 \cdot 2 + 0.7 \cdot 1 = 1.3.$$

Why Relative Performance Matters

Suppose strategy i performs better than average:

$$f_i(x) > \bar{f}(x).$$

Then we expect its share to increase over time.

If instead:

$$f_i(x) < \bar{f}(x),$$

then we expect its share to decrease.

Selection principle

Growth is driven by **relative advantage**.

Replicator Dynamics

The most classical update rule in EGT is the **replicator dynamic**:

$$\dot{x}_i = x_i(f_i(x) - \bar{f}(x)).$$

Here:

- ▶ x_i is the current proportion of the population using strategy i
- ▶ $f_i(x)$ is the payoff of strategy i under population state x
- ▶ $\bar{f}(x)$ is the average payoff in the population
- ▶ $\dot{x}_i$ is the rate of change of the proportion of strategy i

So:

- ▶ if strategy i performs above average, its share increases
- ▶ if strategy i performs below average, its share decreases
- ▶ if it performs exactly at average, its share stays unchanged

Interpretation

The growth of a strategy depends on both its **current prevalence** and its **relative payoff advantage**.

Reading the Replicator Equation

Consider

$$\dot{x}_i = x_i(f_i(x) - \bar{f}(x)).$$

Each term has a simple meaning:

- ▶ x_i : rare strategies change slowly in absolute mass
- ▶ $f_i(x)$: payoff to strategy i
- ▶ $\bar{f}(x)$: population baseline
- ▶ $f_i(x) - \bar{f}(x)$: comparative advantage

Key intuition

Replicator dynamics tracks how **comparative success reshapes the population**.

Matrix Games Under Replicator Dynamics

For a matrix game with payoff matrix A , the expected payoff to strategy i is:

$$f_i(x) = (Ax)_i.$$

The population average payoff is:

$$\bar{f}(x) = x^\top Ax.$$

So the replicator dynamic becomes:

$$\dot{x}_i = x_i((Ax)_i - x^\top Ax).$$

Why this matters

Game payoffs induce concrete population dynamics.

A Running Example: Prisoner's Dilemma

Consider the Prisoner's Dilemma payoff matrix:

$$A = \begin{pmatrix} 3 & 0 \\ 5 & 1 \end{pmatrix}$$

where the two strategies are:

- ▶ Cooperate
- ▶ Defect

To see why defection dominates:

- ▶ if the opponent cooperates: $5 > 3$
- ▶ if the opponent defects: $1 > 0$

Strategic fact

Defection strictly dominates cooperation.

What happens dynamically at the population level?

Checking the Dynamics (Prisoner's Dilemma)

Let x be the fraction of cooperators.

From the payoff matrix:

$$f_C(x) = 3x, \quad f_D(x) = 1 + 4x$$

Compare:

$$f_D(x) - f_C(x) = 1 + x > 0 \quad \text{for all } x \in [0, 1]$$

Key implication

Defectors always have higher payoff than cooperators.

Under replicator dynamics:

$$\dot{x} = x(f_C - \bar{f}) < 0$$

Conclusion

The fraction of cooperators decreases over time.

A Different Example: Coordination

Consider a coordination game such as:

$$A = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}.$$

Here:

- ▶ if many agents play strategy 1, strategy 1 becomes more attractive
- ▶ if many agents play strategy 2, strategy 2 becomes more attractive

Dynamic consequence

The system may converge to different stable conventions depending on the initial population state.

What Kinds of Behaviour Can Emerge?

Replicator dynamics can exhibit several qualitative behaviours:

1. convergence to a pure strategy (e.g., Prisoner's Dilemma $\rightarrow$ all defect)
2. convergence to a mixed population state (e.g., Hawk–Dove equilibrium)
3. selection among multiple stable conventions (e.g., coordination games)
4. coexistence of strategies (stable diversity in the population / stable polymorphism)
5. oscillation or cycling (e.g., Rock–Paper–Scissors)

Main message

Evolutionary analysis studies not only *what is stable*, but also *how populations evolve over time*.

From One Population to Multiple Populations

So far, we have considered a single population state x .

But many strategic systems involve distinct roles or groups, for example:

- ▶ row player vs column player
- ▶ buyers vs sellers
- ▶ planners vs executors

New idea

We may need separate evolving populations, each with its own strategy distribution.

Multi-Population Dynamics

In a two-population setting, let:

x = distribution in population 1, y = distribution in population 2.

Then each population evolves according to its own payoff comparisons:

$$\dot{x}_i = x_i(f_i(x, y) - \bar{f}(x, y)),$$

$$\dot{y}_j = y_j(g_j(x, y) - \bar{g}(x, y)).$$

Interpretation

Each population adapts in response to the strategic environment created by the other.

Why Multi-Population Dynamics Matter

Multi-population models are useful when different roles face different incentives.

They can capture:

- ▶ asymmetric games
- ▶ role-specific adaptation
- ▶ strategic feedback between groups
- ▶ richer dynamic patterns than single-population models

Example intuition

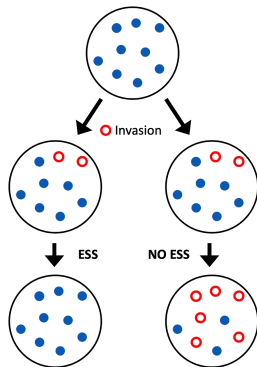
One population may adapt to exploit the current behaviour of the other, which then changes the other's incentives in return.

Evolutionarily Stable Strategy (ESS)

A strategy x^* is an **evolutionarily stable strategy** if it cannot be invaded by a sufficiently small fraction of mutants using an alternative strategy.

Informally:

- ▶ almost everyone plays x^*
- ▶ a small mutant group appears
- ▶ the incumbent strategy still resists replacement



Interpretation

ESS is a notion of **robustness under strategic perturbation**.

ESS vs Nash Equilibrium

These ideas are related, but they are not the same.

- ▶ **Nash equilibrium:** no single player wants to deviate
- ▶ **ESS:** a population using this strategy cannot be invaded by a small group of mutants

So:

- ▶ every ESS is a Nash equilibrium
- ▶ but not every Nash equilibrium is evolutionarily stable

High-level distinction

Nash equilibrium is a **strategic stability** concept.

ESS is an **evolutionary robustness** concept.

High-level lesson

Nash asks whether a strategy is a best response. ESS asks whether it is robust under evolutionary pressure.

What Perspective Has EGT Given Us?

EGT shifts the question from:

“What should a perfectly rational player do?”

to:

“Which behaviours spread, persist, or disappear over time?”

This is especially useful when:

- ▶ adaptation matters more than full rational calculation
- ▶ repeated interaction matters more than one-shot reasoning
- ▶ conventions and norms emerge over time

Why EGT Matters for Multi-Agent Systems

Evolutionary viewpoints help us reason about:

- ▶ behavioural adaptation
- ▶ stability and instability
- ▶ the emergence of conventions
- ▶ strategic selection under feedback
- ▶ cycling and non-convergence

Bridge forward

Today we studied a **population dynamics** view of behaviour. Next, we ask whether **learning algorithms** can also be analysed through a similar dynamical lens.

A Mental Model to Keep

- ▶ Classical game theory: **What strategic outcomes are stable?**
- ▶ EGT: **How do strategy distributions evolve over time?**
- ▶ Multi-agent learning: **How do adaptive decision-makers change through feedback?**

Today's core message:

Evolutionary game theory gives us a dynamic, population-level language for strategic behaviour.

Summary

1. EGT studies how strategy frequencies change over time in a population.
2. Replicator dynamics models growth through relative payoff advantage.
3. Multi-population dynamics captures role-dependent strategic adaptation.
4. ESS provides a robust notion of stability under perturbation.

Looking Ahead

Next lecture:

- ▶ Can reinforcement learning itself be viewed as a dynamical system?
- ▶ How are policy updates related to evolutionary dynamics?
- ▶ Why is this useful for understanding MARL?

Bridge to Lecture 24

If strategy distributions evolve over time in populations, then perhaps **learning algorithms** should also be analysed in terms of the dynamics they induce.